COSC-4117EL: Assignment 3 Report

Group Number: 14

NAME	STUDENT#	EMAIL
Jai Dey	0477282	jdey@laurentian.ca
Seyyedeh Shima Nejati	0461677	snejati@laurentian.ca

Repository URL:
https://github.com/deyjai/Age-Classification-via-Transfer-Learning-on-ResNet18

Problem Statement:

Age classification is the task of figuring out which age group a person belongs to based on an image. Instead of predicting an exact age, it's treated as a multi-class classification problem, where the model assigns each image to one of several age categories. For example, these could be broad groups like child, young adult, adult, middle-aged, and senior. However, in our project, the categories are much more detailed. We split ages from 0 to 80 into 3-year intervals, which gives us 27 different classes.

Age prediction, especially from facial images, is widely studied but still difficult. This is because aging doesn't happen in a simple or consistent way—it's continuous, non-linear, and varies a lot between individuals. Things like ethnicity, gender, skin texture, lighting, facial hair, glasses, and makeup can all affect how old someone looks. Because of this, people who are the same age can look very different, while people of different ages can sometimes look similar. Research by Guo and Mu [2] also showed that age-related features appear at different spatial scales in the face, which makes simpler models less effective. In this work, we aim to address these challenges using ResNet-based architectures.

Applications of Age Classification

Being able to estimate someone's age or age group from an image has several real-world uses across different fields such as:

1. Human-Computer Interaction

   Systems can adjust how they behave based on the user's age. For example, they might use simpler language for younger users, increase font sizes for older users, or change how complex certain information is. Having real-time age detection allows devices like phones or smart assistants to personalize the experience without the user needing to input their age manually.

2. Retail Analytics and Personalisation

Businesses rely heavily on demographics, and age is one of the most important factors. Retailers and advertisers use this information to adjust recommendations, pricing, and marketing. With computer vision, stores can estimate age groups through cameras in real time, allowing them to gather useful insights without needing customers to provide personal information.

3. Forensic Analysis and Law Enforcement

   In forensic situations, estimating age from images helps in cases like identifying victims or analyzing sensitive materials. Moreover, automated age classification can speed things up by notifying investigators of higher-priority cases, particularly those involving potential minors.

Why Deep Learning is the Appropriate Approach

There are several classical methods that have been used for facial recognition tasks, such as Appearance Models (AAM), Local Binary Patterns (LBP), Gabor wavelets, and approaches based on support vector machines or regression models [4]. However, these methods come with clear drawbacks, particularly that they rely heavily on handcrafted feature extraction and often struggle to achieve high accuracy.

Over the past two decades, research has shifted toward using neural network architectures, especially Convolutional Neural Networks (CNNs) [5]. CNNs address many of the limitations of traditional methods by automatically learning hierarchical and task-specific features directly from the data. When trained on large facial datasets, deep learning models consistently outperform older, handcrafted approaches on benchmarks like MORPH, FG-NET, and APPA-REAL [6].

In particular, Kaiming He and colleagues showed that residual networks (ResNets) allow for much deeper models without the usual degradation problems, making them very effective for detailed visual tasks like age estimation [7]. That said, bigger models are not always better. Architectures like ResNet-18 and ResNet-34, while powerful, can be unnecessarily complex for this specific task.

In our case, experimental results showed that a simpler 6-layer network actually performed better than these deeper ResNet models. It achieved a strong balance between accuracy and training time, while the larger models showed diminishing returns. Even though residual connections are designed to reduce issues in deep networks, they didn't fully prevent overfitting in ResNet-34, which ended up lowering its overall performance compared to the smaller ResNet-18 [8].

Our Methodology: Transfer Learning

As it has been stated earlier, age classification is a visual recognition problem. Hence our model needs to understand features such as edges, textures, and facial structures. These features are

already learned from large datasets like ImageNet, so instead of training from scratch, we can reuse this knowledge with a pre-trained ResNet model.

As such, we used the transfer learning approach. Reusing an old model allows our model to converge faster, performs better on smaller datasets, and requires less computational power compared to training a model from the beginning.

We will be using ResNet18 as the base model. This model is chosen because it provides a good balance between accuracy and efficiency. It is deep enough to capture meaningful facial features, but still lightweight enough for faster training and real-time inference. Compared to deeper models like ResNet50 or ResNet101, ResNet18 is more practical for applications such as webcam-based classification.

However, our model ResNet18 is not perfectly designed for image classification, so we have to make a couple modifications:

1. Firstly, we replace the final fully connected layer. The original layer is designed for ImageNet classification, so we modify it so that its output size matches the number of age classes (or age ranges) in our dataset.
2. We freeze the early convolutional layers. These layers already are able to detect corners and edges, so there is no need to perform backpropagation on them. Only the later layers are truly needed to adapt to specific facial patterns. This saves us training time and prevents overfitting.
3. Lastly, we take the output from the final layer and interpret them using softmax and argmax. This allows us to convert raw outputs into probabilities and then select the most likely age class.

<u>Methodology: How does Real-Time Webcam Inference Work?</u>

Firstly, we capture video frames continuously from the webcam using the OpenCV library. This gives us a stream of images that our model will process one by one.

Then for each frame, we resize and preprocess it so that it matches the input format expected by our model (224 × 224 resolution with normalization using ImageNet mean and standard deviation for ResNet18). Our model was trained on data in this exact format, so we need to make sure that every input from the webcam is standardized for accurate results.

Then, we convert the frame into a tensor, to allow computations, so we can run inference on the frame using model.eval(), which sets the model to evaluation mode.

Once the model produces an output, we extract the predicted age class. This is done by applying a softmax (if needed) and then selecting the class with the highest probability using argmax. This gives us the predicted age range.

Finally, we overlay this prediction onto the original frame as text and display it in real time. This entire process repeats for every incoming frame, creating a continuous live prediction system.

To maintain real-time performance (around 15–30 FPS), we reduce computational cost by using smaller frame sizes, skipping frames (for example, only classifying every 2nd or 3rd frame), and optimizing inference using GPU acceleration and lightweight preprocessing.

Modifications needed on our model during evaluation

For webcam classification, our model requires the same adaptations as the ones needed during training. The final fully connected layer needs to adjusted to match the number of output classes followed by converting these outputs into probabilities using softmax and argmax to predict the final age class.

Evaluation Metrics

Firstly, to evaluate the age-classification model, we use several metrics: accuracy, precision, recall, and F1-score. These are reported as both macro and weighted averages. In addition, we also use a normalized confusion matrix to better understand how the model is making predictions.

From the test results, we obtained the following values:

```
Loaded checkpoint from model_state.pth for evaluation-only mode.

Validation Metrics
--------------------------------------------------------------------------
Metric                              Value
--------------------------------------------------------------------------
Accuracy                           0.7355
Macro Precision                    0.7808
Macro Recall                       0.6207
Macro F1                           0.6566
Weighted Precision                 0.7975
Weighted Recall                    0.7355
Weighted F1                        0.7369

Test Metrics
--------------------------------------------------------------------------
Metric                              Value
--------------------------------------------------------------------------
Accuracy                           0.7424
Macro Precision                    0.7809
Macro Recall                       0.6248
Macro F1                           0.6554
Weighted Precision                 0.8014
Weighted Recall                    0.7424
Weighted F1                        0.7406
Metrics saved under: artifacts\metrics
```

Figure 1: Evaluation Table:

*Note that all tests were calculated using a tolerance of 3 bins, which allows us to extend the bin radius from 3 to 9. If the age was a little off from the ground truth, it would still be counted as a correct guess.

Description of Each Measure

Firstly, accuracy gives us an overall measure of correctness. It tells us how many predictions were correct out of all predictions, so it works well as a general performance indicator. An accuracy of 0.7355 is somewhat reliable, but there is a significant be a margin of error.

Then, precision tells us how often the model's predicted age bins are actually correct. This is useful when we want to avoid assigning the wrong age to someone.

Next, recall measures how many actual samples from each age bin are correctly identified. This is especially important for ensuring that smaller or underrepresented age groups are not ignored by the model.

F1-score then balances precision and recall. Because of class imbalance, F1-score is often more informative than accuracy alone.

Additionally, macro averages treat all age bins equally, regardless of how many samples they have. This helps us identify weaker performance in difficult or less frequent age groups.

On the other hand, weighted averages take class frequency into account, so they reflect how the model performs across the dataset as a whole.

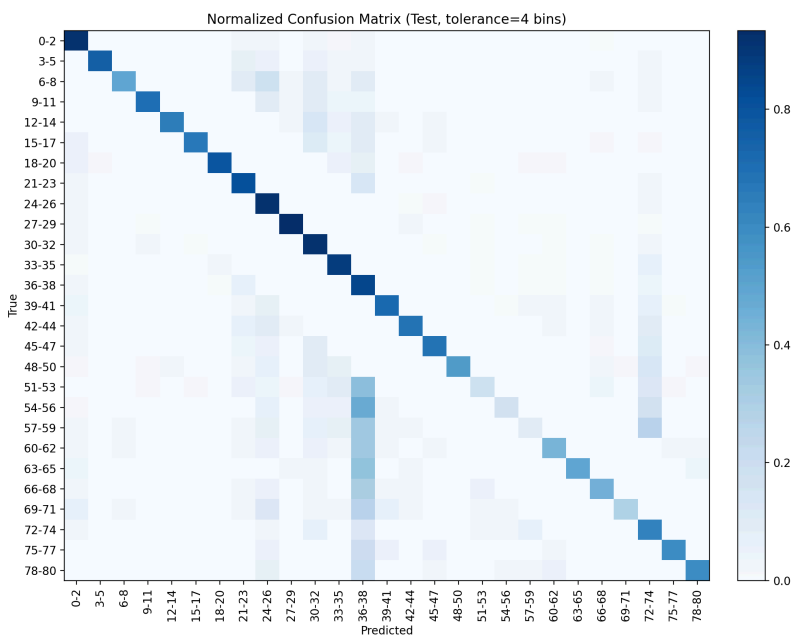

Figure 2: Confusion Matrix

As seen above, the confusion matrix supports these results. It shows a strong diagonal, which means most predictions are correct. Most errors occur between neighboring age bins, which is expected because adjacent age ranges tend to have very similar facial features. The only range of error seemed to be within the 50-60 age group, where the diagonal is the weakest. This could possibly be because this is the "transitionary" phase between adult and senior, and hence many people within this age can look much younger or older than they actually are.

Challenges in Real-Time Webcam Inference

When moving to real-time webcam inference, a few challenges come up as noted below:

1. We need to make sure that FPS stays stable. The FPS can very easily drop if this issue isn't addressed because we are running preprocessing and model inference on every frame. Fortunately, ResNet18 is lightweight. Moreover, we resize frames to a lower resolution, and only run inference every few frames. We also use GPU acceleration with torch.no_grad() and model.eval() to speed things up.

2. Predictions can be a bit unstable between frames. To handle this, we smooth predictions over time, by averaging recent outputs, and keep the previous prediction when confidence is low.

3. Facial expressions and positioning of the user's face needs to be exact in order for the model to be accurate. The dataset UTKface only consists of close-up faces, placed exactly at the middle of the screen with the chin at the bottom and the forehead at the top.

Comparison: ResNet vs YOLO

As stated throughout the report, our model, based on ResNet, takes in a full image and outputs a single age bin (for example, 18–20).

On the other hand, a YOLO-based approach is mainly used for object detection. Instead of just predicting a label, it detects faces in an image and outputs bounding boxes along with predictions. This allows it to handle multiple people in the same frame.

The main difference is that our model predicts one age label for the entire frame, while YOLO can detect multiple faces and assign an age to each one. Because of this, our model works best when there is a single, centered face, while YOLO is better for more complex scenes with multiple people.

In terms of performance, our approach is simpler and faster since it only does classification. YOLO is slightly heavier but still capable of real-time performance on good hardware.

One limitation of our current pipeline is that it does not include face detection. This means predictions can be affected by background noise or multiple faces in the frame. A good improvement would be to first detect faces (using a YOLO-based or lightweight detector), then

crop each face and run our ResNet18 classifier on each one. This keeps our current model while improving accuracy and robustness in real-world scenarios.

Video Demonstration

https://youtu.be/YCO9AYrfjV0

References:

1. P. D. Moral, S. Nowaczyk and S. Pashami, "Why Is Multiclass Classification Hard?," in IEEE Access, vol. 10, pp. 80448-80462, 2022, doi: 10.1109/ACCESS.2022.3192514.

2. Guo, G. and Mu, G. (2013) 'Simultaneous dimensionality reduction and human age estimation via kernel partial least squares regression', Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 657–664.

3. Interpol (2021) Facial Recognition: Uses in Law Enforcement. INTERPOL General Secretariat, Lyon.
4. [1] Lanitis, A., Taylor, C.J. and Cootes, T.F. (2002) 'Toward automatic simulation of aging effects on face images', IEEE Transactions on Pattern Analysis and Machine Intelligence, 24(4), pp. 442–455.

5. G. Levi and T. Hassncer, "Age and gender classification using convolutional neural networks," 2015 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Boston, MA, USA, 2015, pp. 34-42, doi: 10.1109/CVPRW.2015.7301352.

6. Agustsson, E. et al. (2017) 'Apparent and real age estimation in still images with deep residual regressors on APPA-REAL database', Proceedings of the IEEE International Conference on Automatic Face and Gesture Recognition, pp. 87–94.

7. He, K. et al. (2016) 'Deep residual learning for image recognition', Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 770–778.

8. Aydogdu, M. F., Celik, V., and Demirci, M. F., "Comparison of Three Different CNN Architectures for Age Classification", in 2017 IEEE 11th International Conference on Semantic Computing (ICSC), 2017, Art. no. 79. doi:10.1109/ICSC.2017.61.

9. Haq I, Mazhar T, Asif R . YOLO and residual network for colorectal cancer cell detection and counting Heliyon, 2024; 10

10. https://chatgpt.com/s/cd_69e98f2778a8819185237ef1ab7f3055

11. https://chatgpt.com/share/69e9900f-c4d8-83ea-9f58-fabd301d7a34.

12. https://chatgpt.com/s/cd_69e98fc62d8481918fd49f0d37f1c70d

13. https://chatgpt.com/s/cd_69e9905f8c3c8191b4e822d2e4bf2367